

KAUST 2024 OLD Theory Exam

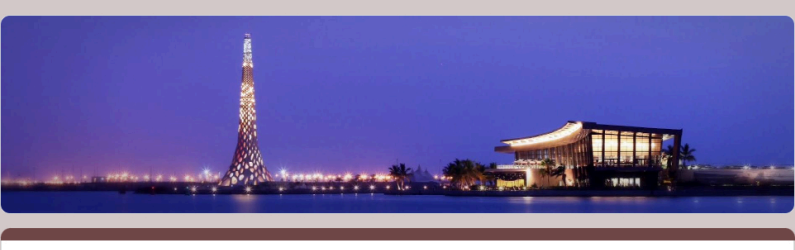
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Madinah Introduction to AI Cohort 2

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* Indicates required question

Introduction to AI Exam

How do you select an initial parameter value for the first iteration of gradient descent?

- ☐ Set it to one
- ☐ It should always be 0
- ☐ Randomly between 0 and 1
- ☒ Randomly ✓

Clear selection

In Numpy, you cannot perform this operation A + B
Given that A's shape is (2, 3) and B's shape is (1, 3)

- ☐ True
- ☒ False ✓

Clear selection

What is the primary goal of hyperparameter tuning in machine learning?

- ☐ Minimizing the number of features
- ☐ Maximizing model interpretability
- ☒ Finding the optimal values for parameters that are not learned during training
- ☐ Reducing the size of the training dataset

Clear selection

In logistic regression, what is the output variable type?

- ☐ Continuous
- ☒ Categorical
- ☐ Binary
- ☐ Both a and b

Clear selection

What is the primary purpose of activation functions in neural networks?

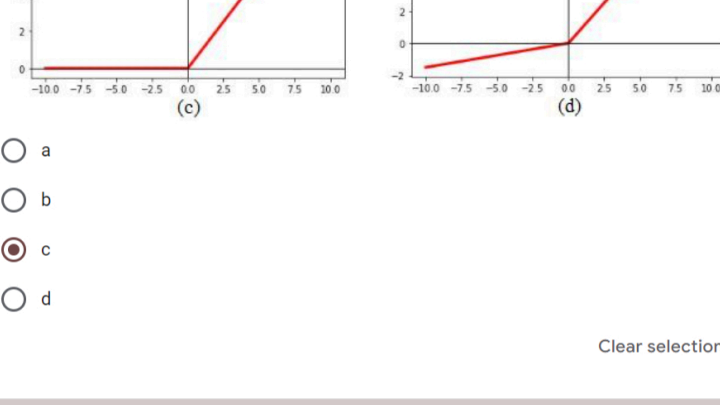
- ☐ Control the learning rate
- ☒ Introduce non-linearity to the model
- ☐ Determine the number of layers
- ☐ Define the number of neurons in a layer

Clear selection

How is the relationship between independent and dependent variables represented in Linear Regression?

- ☒ Linear equation
- ☐ Quadratic equation
- ☐ Exponential equation
- ☐ Logarithmic equation

Which one of these plots represents a ReLU activation function?



- ☐ a
- ☐ b
- ☒ c
- ☐ d

Clear selection

In the softmax function, what is the range of output values?

- ☒ 0 to 1
- ☐ -1 to 1
- ☐ -∞ to ∞
- ☐ 0 to ∞

Clear selection

How is logistic regression different from linear regression?

- ☐ Linear regression is for classification, while logistic regression is for prediction.
- ☒ Linear regression is for prediction, while logistic regression is for classification. ✓
- ☐ Both are the same.
- ☐ Logistic regression is only used for binary outcomes.

Clear selection

Why do we use activation functions, such as ReLU and sigmoid, in neural networks?

- ☐ To reduce the dimensionality of the input data.
- ☐ To handle missing data in the input.
- ☐ To speed up the training process.
- ☒ To introduce non-linearity into the model.

Clear selection

What is the result of the following:

X = torch.tensor([1, 0, 0, 1])
Y = torch.tensor([2, 1, 1, 2])
X_times_Y = X * Y

$$\begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} = 1 \times 0 = 0$$

- ☒ tensor([2, 0, 0, 2])
- ☐ tensor([0, 1, 1, 4])
- ☐ tensor([2, 0, 0, 2])
- ☐ tensor([0, 1, 1, 4])

Clear selection

Which hyperparameter would you change and how, if the training loss of a neural network is always increasing with epochs. (Assuming there are no other errors in the code)

- ☒ Decrease the learning rate
- ☐ Increase the number training epochs
- ☐ Decrease the number of training epochs
- ☐ None of the above

Clear selection

Consider the following output of the lines used in the softmax function shown in the following table. What will be the value of yhat when?

	z_0	z_1	z_2
i	0	1	2
	-3.05	1.46	2.6

$\hat{y} = \text{argmax}_i \{z_i\}$

- ☐ yhat = 2.6
- ☐ yhat = 1
- ☒ yhat = 2
- ☐ yhat = 1.46

Clear selection

What happens when the input to the ReLU activation function is negative?

- ☒ Output is set to zero
- ☐ Output is set to one
- ☐ Output becomes negative
- ☐ Output remains unchanged

Clear selection

What is the primary purpose of cross-entropy in the context of machine learning?

- ☐ Maximize model accuracy
- ☐ Minimize the computational complexity
- ☐ Quantify the difference between predicted and actual probability distributions
- ☐ Normalize input features

What is the primary purpose of feature engineering in machine learning?

- ☐ Minimize model complexity
- ☐ Maximize training time
- ☐ Enhance interpretability of the model
- ☒ Improve the performance of the model by selecting or transforming relevant features

Clear selection

What is the main benefit of using the ReLU activation function in neural networks?

- ☐ Helps prevent the vanishing gradient problem
- ☐ Generates non-smooth outputs
- ☐ Applicable for binary classification exclusively
- ☐ Reduces the risk of overfitting

In which type of problems is the sigmoid function commonly used?

- ☐ Regression
- ☒ Binary classification
- ☐ Multiclass classification
- ☐ Dimensionality reduction

Clear selection

Gradient Descent with Momentum helps with what problems:

- ☐ local minimum
- ☒ vanishing gradients
- ☐ Exploding gradients
- ☐ Global minimum

Clear selection

What is a key difference between binary cross-entropy and categorical cross-entropy?

- ☒ The number of classes
- ☐ The presence of a softmax activation function
- ☐ The range of possible values
- ☐ They are mathematically identical

Clear selection

What is the output of Logistic Regression interpreted as?

- ☒ Probability
- ☐ Residuals
- ☐ Coefficients
- ☐ Variance

Clear selection

What is the activation function commonly used in Logistic ?Regression *

- ☒ Sigmoid
- ☐ ReLU
- ☐ Tanh
- ☐ Softmax

The equation below is called

$$A^T A x = A^T b.$$

- ☐ Normal Equation
- ☐ Regression Equation
- ☐ Gradient Descent
- ☐ Matrix Inversion

In Logistic Regression, the dependent variable is:

- ☒ Continuous
- ☐ Categorical
- ☐ Discrete
- ☐ Ordinal

Clear selection

What type of problem is Logistic Regression commonly used for?

- ☐ Regression
- ☒ Classification
- ☐ Clustering
- ☐ Association

Clear selection

What task does the following lines of code perform?

```
q=torch.tensor(1.0,requires_grad=True)  
fq=2q**3+q  
fq.backward()  
q.grad
```

- ☐ Makes a function that we can use in any part of the code
- ☒ Determines the derivative of 2q**3+q and stores it in q.grad
- ☐ Differentiates the function with respect to all values
- ☐ Performs a forward pass through a neural network

Clear selection

What is the primary purpose of the softmax function in machine learning?

- ☐ Binary classification
- ☐

☒ Multiclass classification

☐ Regression

☐ Dimensionality reduction

Clear selection

? What does the argmax function return

☐ The maximum value in an array

☐ The index of the minimum value in an array

☒ The index of the maximum value in an array

☐ The sum of all array elements

Clear selection

Why do we use bias in the Linear equation?

☐ To introduce randomness and improve model generalization

☐ To reduce the computational complexity of the model

☒ To offset any potential imbalance in the input data

☐ To allow the model to fit better to the training data by shifting the activation function

Clear selection

What does the following line of code do?

`optimizer.step()`

☐ Makes a prediction

☐ Clears the gradient

☐ Makes an update to its parameters

☒ Computes the gradient of the loss with respect to all the learnable parameters

Clear selection

What's wrong with the following code of training model?

for epoch in range(num_epochs):

 outputs = model(X)

 loss = criterion(outputs, y)

 total_loss.append(loss.item())

 loss.backward()

 optimizer.step()

☐ The model object has not been created.

☒ Tensors should be optimized does not exist

☐ There is no loss function

☐ You have to clear the gradient

Clear selection

Consider the following code:

a = np.random.randn(5, 6)

b = np.random.randn(5, 1)

c = a + b

What will be the shape of c?

☒ (5,6)

☐ (5,1)

☐ (10,7)

☐ (0,5)

Clear selection

How does the softmax function normalize the input values?

☐ By taking the logarithm

☐ By applying the exponential function

☒ By dividing each value by the sum of all values

☐ By subtracting the mean

Clear selection

What is the primary goal of Linear Regression?

☒ Minimize Mean Squared Error

☐ Maximize Accuracy

☐ Optimize Precision

☐ Maximize Recall

Clear selection

In a neural network with multiple softmax output nodes, what is the relationship between the probabilities of these nodes?

☐ Independent probabilities

☒ Dependent probabilities

☐ Mutually exclusive probabilities

☐ Constant probabilities

Clear selection

What is correct about stochastic gradient descent?

☐ It does not require a learning rate parameter

☐ It guarantees convergence to the global minimum of the loss function

☐ It's an approximation of batch gradient descent

☐ It requires a fixed learning rate throughout training

In gradient descent, what happens when you select a learning rate that is too large?

☐ It may take too long to converge to a minimum

☒ You may miss the minimum and your loss will start increasing

☐ The loss function will become concave.

☐ The convergence is faster and the loss decreases rapidly

Clear selection

How can we find the minima of a function

☐ Find the first derivative and check if it equals to 1

☐ Check if the Second derivative is positive, given that the first derivative is equal to 0

☐ First the first derivative and check if it equals to 0

☒ Check if the Second derivative is negative given then the first derivative is equal to 0

Clear selection

What is a characteristic of overfitting in machine learning?

☐ High accuracy on both training and new data

☒ High accuracy on training data, low accuracy on new data

☐ Low accuracy on both training and new data

☐ Low accuracy on training data, high accuracy on new data

Clear selection

What does the x-axis typically represent on a loss plot during training iterations?

☒ Epochs

☐ Learning rate

☐ Features

☐ Batch size

Clear selection

What is the purpose of the following line of code?

`x.view(-1,28*28 )`

☐ We would like to view the first 28 elements of a tensor

☒ Convert to vector with 28* 28 elements

☐ We need to multiply every element by 28

☐ Convert images shape to 28 , 28

Clear selection

What is the range of possible values for the dependent variable in Linear Regression ?

☐ [0, 1]

☒ $(-\infty, \infty)$

☐ [1, ∞)

☐ [0, ∞)

Clear selection

How do you calculate the gradient and perform the update for Multiple Linear Regression with 10 input variables?

☐ `loss.backward(d=10) optimizer.step(d=10)`

☒ `loss.backward() optimizer.step(d=10)`

☐ `loss.backward() optimizer.step()`

☐ `loss.backward(d=10) optimizer.step()`

Clear selection

The maximum value of the derivative of the sigmoid function is:

☐ 1

☒ 0.25

☐ 0.1

☐ 0.5

Clear selection

You have 400 samples of data and your batch size is 32. How many iterations will it take to go through 1 epoch?

☐ 12

☒ 13

☐ 32

☐ 23

Clear selection

When the test loss is higher than the train loss this is called

☐ Under-optimization

☒ Over-fitting

☐ Under-fitting

☐ Over-optimization

Clear selection

Consider the following code. How many rows and columns does the tensor yhat contain?

`X=torch.tensor([[[1,0,1,0,1],[1,0,2,0,1],[1,0,3,0,1],[1,0,3,0,1]])`

`model = nn.Linear(3,1)`

`yhat=model(X)`

☐ 4,1

☐ 3,1

☐ 4,2

☐ 4,4

What do both overfitting and underfitting have in common in terms of their impact on predictions?

☐ They improve generalization to new data

☒ They lead to poor predictions on new data

☐ They have no effect on prediction accuracy

☐ They result in perfect predictions on new data

Clear selection

In machine learning, what is the purpose of a loss function?

☐ To maximize accuracy

☐ To minimize computational complexity

☒ To quantify the error between predicted and actual values

☐ To control overfitting

Clear selection

What distinguishes unsupervised learning from supervised learning in machine learning?

☐ Use of labeled training data

☒ Learning from unlabeled data to discover patterns, structures, or relationships

☐ The focus on classification tasks only

☐ Dependence on human-labeled examples

Clear selection

When evaluating a model's performance, which aspect should be considered alongside the loss plot?

☐ Execution time

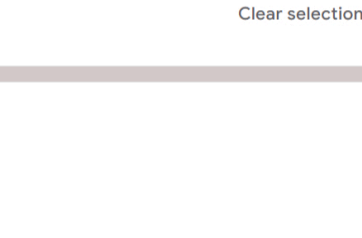
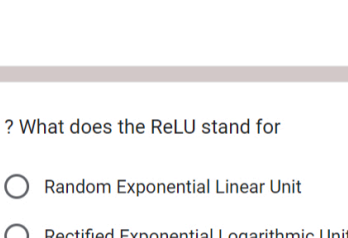
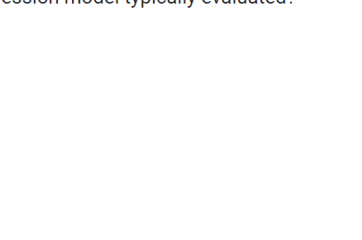
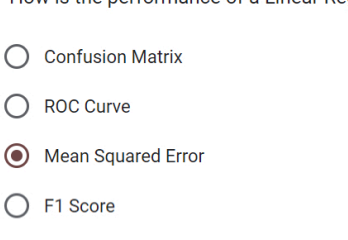
☐ Model architecture complexity

☐ Number of features

☒ Additional metrics like accuracy, precision, or recall

Clear selection

Which one of these plots represents a Sigmoid activation function?



☒ a

☐ b

☐ c

☐ d

Clear selection

How is the performance of a Linear Regression model typically evaluated?

☐ Confusion Matrix

☐ ROC Curve

☒ Mean Squared Error

☐ F1 Score

Clear selection

? What does the ReLU stand for

☐ Random Exponential Linear Unit

☐ Rectified Exponential Logarithmic Unit

☒ Rectified Linear Unit

☐ Randomized Exponential Logarithmic Unit

Clear selection

The loss function depends on what?

☐ The activation function used in the output layer

☒ The difference between predicted and actual values

☐ The learning rate of the optimizer

☐ The number of hidden layers in the neural network

Clear selection

When do we achieve a good-fit ?

- ☐ When train loss is low, and validation loss is higher
- ☐ When both train and validation loss are high
- ☒ When both train and validation loss are low
- ☐ None of the Above

Clear selection

What is the result of the following:

```
a = torch.tensor([1, 2], [0, 1])  
  
b = torch.tensor([0, 1], [1, 0])  
  
a_times_b = a @ b  
  
a_times_b
```

- ☒ tensor([2,1], [1, 0])
- ☐ tensor([0, 1], [1, 2])
- ☐ tensor([2, 1], [1, 0])
- ☐ tensor(2, 0], [1, 0])

Clear selection

The output of the below code
a = [1, 2, 3, 4]
b = a.copy()
a.append(5)
print("a =", a)
print("b =", b)

- ☐ error
- ☒ a = [1, 2, 3, 4, 5] b = [1, 2, 3, 4]2 مغير
- ☐ a = [1, 2, 3, 4, 5] b = [1, 2, 3, 4, 5]
- ☐ a = [1, 2, 3, 4] b = [1, 2, 3, 4]

Clear selection

When is it a great situation where Feature Engineering is recommended in Linear Regression?

- ☐ When the data is linearly separable
- ☒ When the data is not linearly separable
- ☐ When the loss is generally low
- ☐ None of the Above

Clear selection

In which context is the argmax function often used in machine learning?

- ☒ Image processing
- ☐ Natural language processing
- ☐ Model evaluation
- ☐ Clustering

Clear selection

Assume a simple NN model with 3 neurons and inputs= 1,2,3. The weights to the input features are 4,5 and 6 respectively. Assume the activation function is f(x) = 4*x. What will be the output ?

- ☐ 32
- ☐ 64
- ☐ 96
- ☒ 128

Clear selection

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